

Numerical Methods in Chemical Engineering and Problem Solving
ChE 348 (Unique # 17055)
Fall 2026

Course Description: The purpose of this course is to learn basic computational methods for solving a variety of mathematical problems that cannot be solved analytically, and to develop your confidence in using computational tools for problem solving in engineering. The methods will be applied to several problems in chemical engineering. The methods and skills taught in this course will be valuable for future CHE courses (e.g., CHE 372 and CHE 360) and for many numerical solutions you might need to determine in industry and graduate school.

Prerequisites: ChE 210, ChE 317 and M 427K

Lectures: TTh 12:30pm - 1:45pm in CPE 2.218

Recitation: M 1.00pm – 3.00pm in CPE 2.218

Instructor: Dr. Lea Hildebrandt Ruiz lh@che.utexas.edu
Office Hours: Tu 1:45pm – 3.00pm in CPE 5.458, Fri 10.00am -11.00am
on Zoom (see Canvas link) and by appointment

Teaching Assistant: Busra Abdulkadiroglu babdulkadiroglu@utexas.edu
Office Hours: Mo after recitation; Thu 9.30am – 11am EER 5.702

Undergraduate Tutor: Hunter Fugate bhf338@my.utexas.edu
Office Hours: Monday 4-6pm CPE 4.446, Friday 2-3pm CPE 2.222

Course website: on Canvas <http://canvas.utexas.edu>

Required Textbook: None. Recommended textbook: *Numerical Methods with Chemical Engineering Applications*, Kevin D. Dorfman and Prodromos Daoutidis, Cambridge University Press. I will post suggested readings from the recommended text.

Required Computer: Please see the departmental requirement for student computers [[link](#)]. During lecture and recitation, students are expected to have access to a laptop computer with MATLAB.

Assessment:	Individual Homework	60 %
	Midterm 1	20 %
	Midterm 2	20 %

Anticipated course grade cut-offs are A: 93 %, A-: 90%, B+:87%, B:83 %, B-:80%, C+:77%, C: 73%, C-:70%, D+: 67%, D:63%, D-: 60%.
+/- grading will be used

Exams: Midterms 1 will be on October 6; Midterm 2 will be on November 17. There will be no final exam.

Homework: These will be assignments to be worked on your own and uploaded electronically every week on Canvas. Some questions will be simple and guide you through lecture; others will require you to work a problem and write computer code.

COURSE EXPECTATIONS

Knowledge, abilities and skills students should have entering this course:

1. **This course has the following strict prerequisites:** CHE 210, CHE 317 and M 427K with a grade of C at least in each of them. The only exceptions to this policy are students who have already acquired permission from the instructor to waive the prerequisites.
2. Complete understanding of **algebra, trigonometric and logarithmic functions** (Math 305G)
3. Familiarity with **matrices** and their properties.
4. Ability to work with and manipulate **partial derivatives, integration, Taylor series expansions** (Math 408C, 408D, 427).
5. Ability to analytically solve linear **ordinary differential equations of first and second order** (Math 408C, 408D, 427).
6. Understanding of **scientific units, absolute temperature** (CHE 317)
7. Complete understanding of the **behavior of ideal gases** and some familiarity with the **properties of real gases** (CHE 317, CH 302).
8. Fundamental understanding of **mass and energy balances** (CHE 317)
9. **Stoichiometry and equilibrium constants** (Ch 301, 302, 353)

Knowledge, abilities and skills students should gain from this course:

This course has several objectives. The foremost is to introduce several computational techniques which are important in the solution of a variety of mathematical problems that cannot be solved analytically. By the end of this course, students should be able to:

1. Successfully solve mathematical problems with a computer.
2. Be able to formulate and write structured code in MATLAB.
3. Understand the foundation behind the basic numerical methods for matrix manipulations.
4. Solve sets of linear and nonlinear equations using numerical methods as well as built-in MATLAB functions.
5. Apply numerical methods and MATLAB functions to differentiate and integrate a function or a set of discrete points.
6. Apply explicit and implicit numerical methods and MATLAB functions to integrate single and multiple sets of initial value problems.
7. Apply shooting and finite difference methods to solve boundary value problems.
8. Apply finite difference and implicit methods to solve parabolic and elliptic partial differential equations.
9. Apply numerical methods and MATLAB functions to perform optimization in the context of parameter estimation.

The programming environment incorporated into MATLAB will be used to introduce structured programming techniques. The major concepts taught in this course are broadly transferable to other software or computer languages.

TENTATIVE COURSE SCHEDULE

***This syllabus represents my current plans and objectives. As we go through the semester, those plans may need to change. Such changes are not unusual and will be communicated clearly.*

Reading from the recommended text is indicated for each lecture or set of lectures.

#	Date	Topic	Recommended Reading
1	8/25	Introduction; Overview of course; MATLAB	
2	8/27	MATLAB basics	A1-A4, 1.3, 1.4 A5-A8
		Systems of Linear Equations	
3	9/1	HW1 due Introduction to systems of linear equations Application: Mass balance on a simple separator Matrix algebra and properties	2.1-2.4
4	9/3	Gaussian Elimination with pivoting LU decomposition	2.5-2.6 2.11-2.12
5	9/8	HW 2 due Tridiagonal Matrices Application: Liquid-liquid extractor	2.9-2.10
6	9/10	<i>Class canceled for EXPO</i>	
		Non-linear Algebraic Equations	
7	9/15	HW 3 due Review of Taylor series expansion Introduction to Non-linear equations Unbounded methods: Successive Substitution	3.1 – 3.2
8	9/17	Unbounded methods: Newton and Secant Methods Bounded methods continued: Bisection, Regula Falsi Method	3.7-3.8 3.3-3.4
9	9/22	HW 4 due Illinois Method. Systems of Non-Linear Algebraic Equations Application: Multiple reactions in a CSTR	
10	9/24	Numerical Integration and Interpolation	8.1-8.4

#	Date	Topic	Recommended Reading
		Ordinary Differential Equations – Initial Value Problems	
11	9/29	HW 5 due Explicit and implicit Euler Method and numerical stability	4.1-4.2
12	10/1	Higher order and predictor – corrector methods	4.3-4.5
13	10/6	Midterm 1: Matlab, Linear Eqs, Nonlinear Eqs, Numerical Integration and Interpolation (HW1-5)	
14	10/8	Runge-Kutta methods	
		Ordinary Differential Equations (ODEs): Boundary Value Problems	
15	10/13	HW 6 due Introduction to BVPs, Finite difference method Application: reaction/diffusion in a catalytic disk Derivative boundary conditions and other variations	6.1-6.4
16	10/15	Shooting method Application: reaction/diffusion in a catalytic pellet	
17	10/20	HW 7 due Systems of BVPs	6.5
		Partial Differential Equations (PDEs)	
18	10/22	Explicit Central Finite Difference Method Application: unsteady reaction/diffusion in a catalytic disk (Parabolic PDEs)	7.1-7.3
19	10/27	HW 8 due Stability and Implicit Methods (Parabolic PDEs)	
20	10/29	Elliptic PDEs: Heat conduction in a rectangular plate	7.4
21	11/3	HW 9 due PDEs: approximation methods	
22	11/5	TBD	
		Optimization	
23	11/10	HW 10 due Introduction to Optimization	
24	11/12	One Dimensional Search	
24	11/17	Midterm 2: ODEs and PDEs	
25	11/19	Monte Carlo Simulation	

#	Date	Topic	Recommended Reading
26	12/1	Multi-Dimensional Search, Parameter estimation	
27	12/3	HW 11 due	

FEEDBACK

It is my goal to make this an excellent course and a positive experience for all of us. If you feel that the course is not meeting your expectations or you want to provide feedback on how the course is progressing for you, please contact me and/or the TA, either directly or via an anonymous survey in Canvas. I will also be circulating an early course evaluation in class. I will use your constructive comments as a way to improve the course where possible.

COURSE STRUCTURE AND EVALUATION

Homework: Due dates are noted on the course schedule. You must work on these on your own. You may consult your notes, book and even the internet – but you may not copy and paste code from anybody or any resource. We may utilize software designed to detect plagiarism.

Exams: Two closed-book midterm exams will test your conceptual understanding of numerical methods as well as your understanding of MATLAB coding. You will not write code in MATLAB for these exams but may be asked to evaluate, comment on and/or fix code – we will discuss example problems in class.

Recitations: The teaching assistant will conduct recitations where MATLAB syntax and coding will be reviewed and/or a problem similar to problems on the HWs will be given to be solved. This is an excellent opportunity to get advice while you are working on a computer. Please go to recitation and take advantage of it.

POLICIES

Academic Accommodations: Students with disabilities may request appropriate academic accommodations from the Division of Student Affairs, Disability and Access (D&A) – see the [University Policies and Resources for Students](#) Canvas page or visit the [D&A website](#).

Academic Dishonesty: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community. Academic dishonesty encompasses such behaviors as plagiarism, copying another student's work, and a variety of other academically dishonest behaviors. Since such dishonesty harms the individual, all students, and the integrity of the University, all academic dishonesty cases will be referred to Student Conduct and Academic Integrity in the Office of the Dean of Students for resolution.

Artificial Intelligence Policy Statement: In accordance with the University's Institutional Rules on Student Services and Activities, Chapter 11, students accept the responsibility to always uphold academic integrity and an honor code reflective of a scholarly community devoted to academic and personal success. All members of the University community are fully accountable and responsible for any output they produce as part of academic work. They are also responsible for following the guidance specified in the [Texas Statement on Academic Integrity](#) and avoiding prohibited uses of generative AI tools outlined in the Information Security Office's guidance on [Acceptable Use of Generative AI Tools](#). In this course, **generative AI use shall be permitted on a partial basis and only when specifically stated on the assignment.** Using generative AI tools without authorization or failing to disclose generative AI use according to the disclosure policy in this course, even when AI use is permitted, may constitute academic misconduct under UT Austin's Institutional Rules and will be referred to Student Conduct and Academic Integrity in the Office of the Dean of Students for resolution.

Classroom Recording Statement: Students must be informed that course recording is not permitted without faculty permission. Faculty are required to include the following statement on their syllabi to fulfill this requirement: [HOP 2-9970](#) prohibits students from recording class instruction (audio or video) unless a student obtains the instructor's permission or Disability & Access has approved audio recording as an accommodation.

Important Safety Information: If you have concerns about the safety or behavior of fellow students, TAs or Professors, call BCAL (the Behavior Concerns Advice Line): 512-232-5050. Your call can be anonymous. If something doesn't feel right – it probably isn't. Trust your instincts and share your concerns.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture notes, videos, assessments, in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating.

Statement on Flexibility: If you experience any hardships such as illness, accident, family crisis please know deadlines can be amended. Please communicate with me as soon as you feel comfortable doing so. If for any reason you do not feel comfortable discussing with me, please visit Student Emergency Services.

Title IX Reporting: Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating.
2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation.
3. Investigate and discipline violations of the university's relevant policies

Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual

assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. **I am a Responsible Employee and must report any Title IX related incidents** that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

Religious holidays: By UT Austin policy, notify the instructor of your pending absence at least fourteen days prior to the date of observance of a religious holy day. If you miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

University policies and resources: For a list of important university policies and helpful resources that you may need as you engage with and navigate your courses and the university, see the [University Policies and Resources for Students](#) Canvas page. The page includes important safety information including the policy on carrying of handguns on campus, the language of the University Honor Code, Title IX legal requirements for Texas employees, and information about how to receive support through the office of Disability & Access, tutoring services, the Counseling and Mental Health Center, University Health Services, and other resources.

Appendix: Instructions to submit the required files for ChE 348 on Canvas

File and Name Format

Include all files in a zipped folder. Name the folder as follows.

<EID of student>_<Assignment name>

Example: If your EID is abc123 and you want to submit the script files for assignment Coding0119 the name of the zipped folder should be

abc123_Coding0119

Do *not* change the name of the files inside of the folder.

Submission through Canvas

The folder containing the required script files has to be submitted through Canvas

1. Login to Canvas
2. Select the course - ChE 348 - Numerical Methods in ChE and Problem Solving
3. Click on Assignments tab
4. Click on the assignment for which you are submitting files
5. Click on **Submit Assignment** on the right
6. Under **File Upload** browse and choose the file (zipped folder) to attach.
7. After choosing the file it will appear as a selected file.
8. Then click **Submit Assignment** to upload the file.